

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Tech. Examination (EE)

Nov-Dec 2013

EE 302.01 ELCTRICAL POWER SYSTEM-II

Date: 25.11.2011, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 What do you mean by Stringing of transmission line? Explain the Effect of [07]
Temperature, Unstretched length of conductor and Temperature-Sag and
Temperature-Tension curve on it.
- Q - 2 (a) How the Choice of Voltage is linked with the Line Loading and Length of the [06]
Transmission line?
- (b) Explain the Methods of Reducing Corona. Also Give the advantages and [05]
Disadvantages of Corona.
- (c) Explain the Relation between Sag, Span and Tension in Relation with the [03]
Transmission line.

OR

- Q - 2 (a) Define Insulation Co-ordination. With adopted Basic Insulation Levels, How BIL is [06]
important to protect the equipments?
- (b) Explain the following points in Selection of Sizes and Location of Generating [05]
Stations.
- (1) Size of Units (2) Location of Power Stations.
- (c) Define Radio Interference and explain the Propagation of it. [03]

Q - 3 (a) By using following Ratings of Substation, discuss the design criterion of Substation [07]
equipments with necessary ratings.

Incoming Line:- 11 kV, No. of Lines = 3

Outgoing Line:- 220 kV, No. of Lines = 2

440 kV, No. of Lines = 2

Also Give the Location of this Substation.

(b) Give the comparison between Underground system and Overhead system. [05]

(c) Give the advantages of Interconnection. [02]

OR

Q - 3 (a) Derive the equation of minimum potential with Uniformly Loaded Distributor Fed at both ends with equal Voltages. [07]

(b) A single phase a.c. distributor AC 2000 metres long is fed from end A is loaded as under: [05]

(i) 100 A at 0.8 p.f. leading 1000 m from point A

(ii) 100 A at unity p.f. 2000 m from point A

The load resistance and reactance (go and return) as 0.06 and 0.1 ohm/km. The voltage at the farther end is 220 volt. Find the voltage at the sending end, if power factors of the loads are with reference to father-end voltage

(c) Explain the design considerations in Feeders and Distributors for Distribution system. [02]

SECTION – II

Q - 4 (a) Explain the different methods used for forecasts of future demand of electrical energy. [07]

Q - 5 (a) Explain Transmission system planning model with necessary flow diagram. [07]

(b) A 3-core, 11 kV cable supplies a load of 1500 kW at 0.8 pf lagging for 300 days in a year at an average of 8 hours per day. The capital cost per km of cable is Rs 8,000 + [07]

20,000 a ('a' is cross sectional area of the cable). The resistance per km of a cable of cross sectional area 1 cm^2 is 0.173Ω . If the energy loss cost is 2 paise per unit, and the rate of interest and depreciation is 12 %, calculate the most economical current density and diameter of the conductor.

OR

- Q - 5 (a) Explain the Kelvin's Law to determine the size of conductor with its practical [07]
Limitation.
- (b) A two conductor street main AB, 500 meters in length is fed from both ends at 250 [07]
volts. Loads of 50 A, 60 A, 40 A and 30 A are tapped at distances of 100 m, 250 m,
350 m and 400 m from A, respectively. If the cross section area of the conductors be
 1 cm^2 and specific resistance of the material of the conductor is $1.7 \times 10^{-8} \Omega\text{-m}$,
determine the minimum consumer voltage.
- Q - 6 (a) Give the comparison between 1- phase to neutral and Single Wire Earth Return [07]
systems of Rural Supply. Amongst them Which system is more suitable in hilly area
and why?
- (b) Explain the different steps for power system Planning with necessary diagram. [05]
- (c) Give the classification of Distribution system. [02]

OR

- Q - 6 (a) Discuss faults and existing protection systems of Rural Electrification. Which [07]
corrective steps should be required to improve the existing system?
- (b) Explain the Effect of voltage on Transmission efficiency OR Advantages of [05]
Transmitting power over transmission lines on high voltages.
- (c) List out the Application of Auto-Recloser in power system. [02]

BEST OF LUCK
